

Helios Compute Control — illustrative venture investment committee memorandum

SYNTHETIC — NOT INVESTMENT ADVICE

Conditional posture: INVEST subject to the executable terms and falsifiers below.

Authority: DECISION_RECORD_INCOMPLETE / PENDING_FOUNDER_SIGNATURE / no delegated investment authority.

Knowledge cutoff: 2026-08-29T00:00:00Z

Packet receipt: aa69df6565a8473ff0e9e673b76b9970b2537a737e7e9277931ed0ed33fc19fa

Recommendation and executable terms

Invest at the proposed valuation only with milestone-based funding tied to ordinary-cohort retention, verified pipeline conversion, and gross-margin progression.

TERM	EXACT SELECTED-CASE POSITION
Security	Series C, 1x non-participating preference, senior to A/B and junior to any declared bridge
Initial / conditional capital	\$25M at close / \$15M at month 12
Valuation	\$160M pre-money / \$200M fully funded post-money
Series C ownership	13.51% at first close / 20.00% fully funded
Option pool	1,611,031 unissued shares after refresh; pre-money holders bear the refresh
Base returns	47.62% gross XIRR / 7.02x gross MOIC
Down-round returns	8.70% gross XIRR / 1.52x gross MOIC
Shortfall case	Bridge at month 36; 19.00% / 2.39x

Product, market, customers, competition, and business model

Helios can become the system of control for volatile enterprise GPU spend if ordinary cohorts retain and optimizer savings translate into durable platform economics.

DIMENSION	UNDERWRITING JUDGMENT	DISCONFIRMING FACT / OPEN GATE
Product	GPU-spend control plane with customer-level usage and cost records.	Production replication of the optimizer effect remains open.
Market	Tiered Bayesian survey supports a scenario range, not a market fact.	Tier 5 is data-thin and abstained; priors remain explicit.
Customers	Ordinary cohorts are separated from hand-picked design partners.	Pooled retention overstates repeatability.
Competition	Workflow value depends on neutral provider-cost control.	Provider-native tooling and cloud concentration may compress differentiation.
Business model	Usage-linked expansion can compound with customer compute spend.	Gross margin remains exposed to compute, telemetry, and support costs.
Terms	Milestone financing limits capital at risk before operating proof.	Failure creates explicit down-round or bridge dilution.

Counterthesis: Design-partner selection, inflated pipeline, cloud-cost exposure, and preference-heavy outcomes may make growth and TAM appear more durable than they are.

Team and financing accountability

These are role-level evidence states, not invented biographies or reference-checked conclusions.

DIMENSION	ASSESSMENT
Observable strengths	CEO / founder: Product insight is supported by design-partner problem definition. CTO: Optimizer experiment shows synthetic unit-cost improvement. CRO / revenue: Pipeline stage history exists at opportunity level. Finance: No dedicated capability is evidenced.
Unproven capabilities	CEO / founder: Commercial dependence and succession remain open. Evidence state: OPEN. CTO: Provider concentration and production replication remain open. Evidence state: PARTIAL. CRO / revenue: Ordinary-customer repeatability is not established. Evidence state: OPEN. Finance: Usage margin, runway, and financing controls need an accountable owner. Evidence state: ABSENT.
Key-person risk	OPEN — founder commercial dependence and succession evidence remain conditions to the second tranche.
Required capacity	Finance leader accountable for usage margin, monthly close, runway, and financing controls. Revenue operations owner accountable for ordinary-customer stage-to-close governance.

Cap table and financing-event bridge

SCENARIO	FUNDED SERIES C	OWNERSHIP	MINIMUM CASH	EXIT PROCEEDS	GROSS XIRR	GROSS MOIC
BASE	\$25M	12.94%	\$13.4M	\$175.4M	47.62%	7.02x
MILESTONE	\$40M	20.00%	\$17M	\$360M	59.54%	9.00x
DOWNSIDE	\$25M	10.14%	\$9M	\$37.9M	8.70%	1.52x
FINANCING_SHORTFALL	\$25M	10.79%	\$250,000	\$59.7M	19.00%	2.39x

SELECTED EVENT	MONTH	STATUS	CASH	NEW SHARES	FULLY DILUTED SHARES
series-c-close	1	FUNDED	\$25M	1,814,223	13,425,254
series-c-tranche	12	FUNDED	\$15M	1,088,534	14,513,788

Milestone financing and monthly runway

The \$15M second tranche releases only when `ALL_TESTS_PASS_AFTER_CURE` after a 30-day cure. Evaluator: `BOARD_FINANCE_COMMITTEE`. Failure consequence: `WITHHOLD_TRANCHE_AND_REUNDERWRITE_RUNWAY`.

- `hx-nrr-metric` `>= 1.05` over `trailing_12_months`.
- `hx-margin-metric` `>= 0.70` over `trailing_3_months`.
- `hx-pipeline-audit` `== COMPLETE` over `as_of_test`.
- `hx-optimizer-replication` `== REPLICATED` over `pre_tranche`.

YEAR	ENDING CASH
1	\$48.2M
2	\$38.5M
3	\$30.2M
4	\$23M
5	\$17M

Preference waterfall and investor return bridge

Exit value basis is `EQUITY_VALUE`. Unissued pool shares receive zero proceeds. Class proceeds conserve to \$1,600M with a 0-cent residual.

CLASS	ELECTION	PREFERENCE	RESIDUAL	TOTAL PROCEEDS
SERIES_A	CONVERT	\$0	\$254.2M	\$254.2M
SERIES_B	CONVERT	\$0	\$229.4M	\$229.4M
SERIES_C	CONVERT	\$0	\$360M	\$360M

Series C invests \$40M in dated funded tranches and receives \$360M at exit: **59.54% gross XIRR / 9.00x gross MOIC**.

Conditional distribution and sensitivity

The 1,000 retained paths replay the full financing ledger and exact waterfall. They are scenario priors, not a forecast or evidence of real-world accuracy.

STATISTIC	P10	P50	P90
Gross MOIC	1.46x	6.85x	11.47x
Gross XIRR	8.23%	49.51%	71.35%

- Probability below 1.0x: **0.10%**.
- Sensitivity book: 11 full-engine cells across exit value, exit date, later-round price, and milestone state.

Econometric credit and zero-credit map

- HX-06 (CAUSAL_SYNTHETIC_ONLY) supports only the randomized synthetic optimizer estimand and tested population.
- HX-07 (NOT_IDENTIFIED) receives zero base-case causal credit because adoption follows spend spikes and identification fails.
- HX-05 (PREDICTIVE_ASSOCIATION) is a prior-sensitive market-size scenario, not causal evidence.

Value creation and board cadence

Combined full-model impact: **\$3.2M minimum cash, \$3.9M Series C proceeds, 0.36% gross XIRR**, including an explicit **\$0 interaction residual**.

INITIATIVE	EVIDENCE CLASS	BASELINE / TARGET	OWNER	MODELED CONSEQUENCE	STOP RULE / RISK
Ordinary-cohort expansion	HUMAN_JUDGMENT	117.07% / 125%	CRO	\$75,504/month cash; \$9.1M exit value; \$1.5M cost	Design-partner tactics do not transfer; stop if two ordinary cohorts fall below 105%
Optimizer unit economics	MIXED_CAUSAL_SYNTHETIC_AND_SCENARIO	70.61% / 74%	CTO	\$56,437/month cash; \$8.1M exit value; \$2M cost	Provider price changes; stop if replication interval crosses zero
Enterprise sales governance	DESCRIPTIVE	48 inflated opportunities / <15% forecast error	CRO / Finance	\$0/month cash; \$0 exit value; \$1.2M cost	Enterprise cycle elongation; stop tranche if stage history is incomplete

Economic mapping register

- **Ordinary-cohort expansion:** $\text{ordinary_base_arr_cents} * \max(0, \text{target_nrr} - \text{observed_nrr}) * \text{gross_margin} * 50\% \text{ realization}$. Inputs: $\text{exit_multiple_on_annual_cash}=10.0$; $\text{gross_margin}=0.7061223762551445$; $\text{observed_nrr}=1.1706504793654182$; $\text{ordinary_base_arr_cents}=3234114396$; $\text{realization_factor}=0.50$; $\text{target_nrr}=1.25$
- **Optimizer unit economics:** $\text{ltm_compute_cost_cents} * \text{abs}(\text{adjusted_optimizer_itt}) * \text{adoption_rate}$. Inputs: $\text{adjusted_optimizer_itt}=-0.09154958494336665$; $\text{adoption_rate}=0.65$; $\text{causal_boundary}=\text{Synthetic ITT identifies the planted test population only}$; adoption and valuation multiple remain scenario judgments.; $\text{exit_multiple_on_annual_cash}=12.0$; $\text{ltm_compute_cost_cents}=1138084029$
- **Enterprise sales governance:** $\text{zero_base_case_credit_until_stage_history_and_conversion_design_are_identified}$. Inputs: $\text{modeled_value_credit_cents}=0$

PHASE	TIMING	OWNER	MILESTONE	KPI	STOP RULE
Pre-close	Before signing	Deal lead	Verify cap table, preferences, side letters, and milestone definitions.	All financing operands reconciled	Do not fund with unresolved ownership or waterfall terms.
Day 1	Funding date	CEO / CFO	Lock usage-margin, runway, and ordinary-cohort definitions.	Signed operating metric dictionary	Do not release milestone capital on non-reconciled metrics.
Day 30	Close + 30 days	CFO	Install weekly runway and provider-cost forecast.	Cash, committed spend, gross margin	Escalate if runway falls below financing plan.
Day 100	Close + 100 days	CRO / CTO	Reconcile pipeline history and optimizer adoption economics.	Ordinary-cohort NRR, conversion, cost savings	Hold second tranche if milestones are not independently evidenced.
Year 1	Quarterly through month 12	Board	Re-underwrite ownership, runway, and exit cases.	Dilution, runway, margin, milestone state	Replan financing before the declared runway floor.

Falsifiers, open diligence, and limitations

- FALSIFIER: Ordinary-cohort NRR below 100%
- FALSIFIER: Pipeline conversion below 20%
- FALSIFIER: Gross margin below 65%
- FALSIFIER: Runway below 12 months post-close
- OPEN: Full stage-history export
- OPEN: Design-partner contract sample
- OPEN: Cloud-cost unit ledger
- OPEN: Preference and pro-rata side letters

This packet is generated from a fictional deterministic room. Synthetic causal estimates recover planted mechanisms only. Scenario outputs are not forecasts. The investment record remains unsigned and requires human IC authority.

Receipt appendix

- Case analysis: [dc67fce3adecef8ccb32654d3a0384fbf301903d5704ff024ecbb747ce7405c7](#)
- BASE result: [25e42585f6c9cd97a030d3d79dcb69c9aa70795ec76da82e822b8e2a6f0f3417](#)
- MILESTONE result: [9cf24ca939951b9d65eecd0ce26b2721d39f4f0c48578e6a0a9093cbcf2203e8](#)
- DOWNSIDE result: [8c4a2cfe0cafaa5a6340c8f0419f32e97ca590eab1ef8ac25cac9576affe2d20](#)
- FINANCING_SHORTFALL result: [12e20a1bd48af041b1079449c086fe22fb0272ddc7350bba5a0e8f45d43676d0](#)
- Distribution: [313c65dba75bbb7f41df051d6863d0ce505b35667cda0b8e6d762b685ab6c3fb](#)
- Sensitivity book: [2a22c9c3cb2cd91ee537e0cca6ae8081b700e8d778e2dd1e03d9cfc4757b060d](#)
- Value-creation bridge: [ffb804f3e4c8e0b310ddb656c0911b6f55f1da79d7669517799c9edc02b1c08c](#)